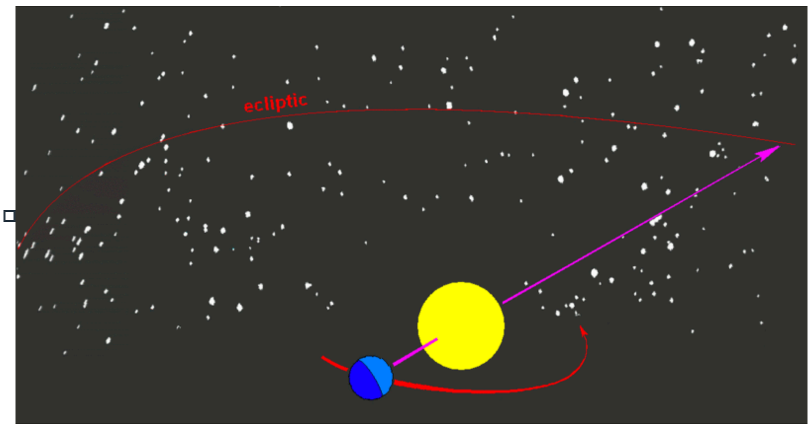
assignment5 celestial sphere , telescopes - bonus ahead

This is a preview of the published version of the quiz

Started: Mar 22 at 7:27pm

Quiz Instructions

2 attempts. last grade counts



Question 1 4.5 pts

The rays coming from a far away objects are parallel

- false
- true
- Idk

Question 2 4.5 pts

Venus/mercury can be seen (unit celestial sphere) - see last slide of unit celestial sphere

- any time of the night
- at midnight
- at sunset or sunrise
- only at sunset

Question 3 4.5 pts

The polygon that includes stars like Sirius, Procyon, Capella, Aldebarran ...

and that can be seen during the Winter is called the Winter ...

- parallelogram
- hexagon
- triangle
- pentagon

Question 4 4.5 pts

In class, you looked through a toy telescope. The image of a far away object can be displayed on a screen . What is true? (what did you see)

you can check here:

https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html (https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html)

move the pencil (left of the lens) far @ left. Compare this object to its image (right side of the lens).

- image was upside down and larger than the object
- image was upright (not upside down) and smaller than the object
- image was upright and larger than the object
- image was upside down and smaller than the object

Question 5 4.5 pts

See my slides (celestial sphere). Orion can be seen above the head (zenith) in (North hemisphere) in

- Mars
- December
- June

September

Question 6 4.5 pts

The position of polaris above the horizon gives us our

longitude

season

latitude

eastern time

Question 7 4.5 pts

At the equator, Polaris is seen (slides celestial sphere, lectures)

above head (zenith)

I depends on the season

at the horizon

between the horizon and the zenith

Question 8 4.5 pts

The planets can be observed

only along the horizon

all over the celestial sphere

only above head at midnight

only along the ecliptic line

Question 9 4.5 pts

The twelve constellations of the zodiac lie along the ecliptic line

Idk

FAlse

True

It depends on the hemisphere

Question 10 4.5 pts

At the North pole, the stars rotate about

polaris

rigel

sirius

altair

Question 11 4.5 pts

If the resolution is poor, 2 stars will appear as one (the images merge together). This is because of

You can wait for the lecture.

refraction (bending of light)

saturation

diffraction (spreading of light)

reflection (bouncing of light)

Question 12 4.5 pts

The eye has a diameter of 6mm. This is 1000 times smaller than a 6 m telescope. So an image will be

hint: area is proportional to diameter squared. You can wait for the lecture

Idk

1,000 dimmer

1,000,000 times dimmer

same brightness

Question 13 4.5 pts

planets don't twinkle as much as stars because (we watched a video in class).

Wait for the lecture

- They don't burn fuel inside
- They don't emit radiation
- they are closer
- they are rocky

Question 14 4.5 pts

Radio telescopes have to be larger than visible light telescope because

- the waves they collect have smaller wavelength
- the waves they collect have larger wavelength
- The size does not matter
- they are shaped like a dish

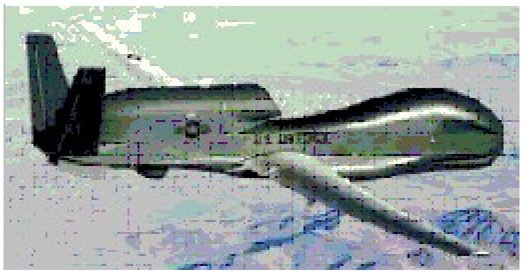
Question 15 4.5 pts

The Global Hawk spy drone flies at an altitude of 19,800 m (19,8km). Its optical sensor (MS-177 camera) has a size of about 0.03 m (30cm). What is the size of the smallest object it can see. (so the resolution). Suppose the wavelength (visible) is 0.0000005 (0.5 micrometer). Here is the Math:
resolution =(wavelength x distance)/ size

so 1) multiply wavelength by distance

2) . and divide the result by the size.

3) Then to get cm multiply by 100.   10cm is about 4in FYI



- 7
- 22
- 33
- 15

Question 16 4.5 pts

You have a better resolution with blue light, red light or radio wave? (when using a optical instrument)

- Its the same resolution
- blue light
- radio wave
- red light

Question 17 4.5 pts

Suppose the angular separation of two stars is smaller than the angular resolution of your eyes. How will the stars appear to your eyes? (meaning what happens if the resolution is bad)

- You will see two distinct stars.
- You will see only the larger of the two stars, not the smaller one.
- You will not be able to see these two stars at all.
- The two stars will look like a single point of light.
- The two stars will appear to be touching, looking rather like a small dumbbell.

Question 18 4.5 pts

The angular resolution of an 8 inch diameter telescope is ________ greater than that of a 2-inch diameter telescope. hint: resolution is proportional to the size of telescopes. so what is 8/2 ?

- 8×
- 4x

- 1×
- 2×

Question 19 4.5 pts

What is the benefit of a longer exposure time when making an image? hint: you wait more time and collect more photons. Light is made of photons (particles of light)

- It allows the detector to measure rapid variations in light intensity.
- It allows the detector to collect more total light.
- It allows the detector to record light at very short wavelengths.
- It allows the detector to achieve higher angular resolution.

Question 20 4.5 pts

Sound, light, water waves all bend around corner. When waves go through openings, they spread out. This is called

- diffraction
- simulation
- refraction
- reflection

Question 21 4.5 pts

The twinkling of stars is due to the ____________- of light as it goes through the layers of the atmosphere (see video we watched)

- refraction
- reflection
- difraction
- simulation

Question 22 4.5 pts

The VLA stands for

- Very Last Aim
- Very Large Array
- Vast Long Acres
- Vast Lonely Asset

Question 23 4 pts

slide 15 (celestial sphere). The stars in the constellation Cassiopea are close to each other (bound by gravity)

- I missed the class
- I spaced out during the class
- False
- True

Question 24 4 pts

At the spring equinox in the Northern Hemisphere, which typically occurs around March 20th or 21st, the sun is in the zodiac constellation of Pisces (Pieces) . But because of the precession of the EArth, in years from now , the sun will be in (remember the song)

- ophiochus
- capricon
- leo
- aquarius

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